



AI-BASED DDOS DETECTION & AUTOMATIC PREVENTION

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA0707 – COMPUTER NETWORKS

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

Submitted by

Avinash M

(192521196)

Course Coordinator

Dr.S.SIVAGAMI

Professor, Department of Data Vista / CSE,
SIMATS Engineering

Course Faculty

Dr.SHANKAR R.

Professor, Department of Quantum
Communication,
SIMATS Engineering

SIMATS ENGINEERING

Saveetha Institute of Medical and
Technical Sciences Chennai-602105

SEPTEMBER 2026

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “AI-Based DDoS Detection & Automatic Prevention” has been carried out by **Avinash M (192521196)**, for CSA0707 – Computer Networks at SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai. The project work presented in this report describes the software-based design, implementation, testing and evaluation of an AI-assisted DDoS detection and automatic prevention prototype developed in a controlled local simulation environment.

The work combines network traffic simulation and monitoring, feature extraction, Random Forest based classification, temporary source blocking and a Security Operations Center (SOC)-style dashboard. The supplied project presentation records the implementation and its observed system-level behaviour, including automatic traffic classification and temporary blocking without manual intervention.

COURSE COORDINATOR

Dr.S.SIVAGAMI

Professor,

Department of

Data vists / CSE

SIMATS Engineering,
SIMATS,Chennai - 602105

COURSE FACULTY

Dr.SHANKAR.R

Professor,

Department of

Quantam Communication

SIMATS Engineering,
SIMATS,Chennai - 602105

DECLARATION

I, **Avinash M (192521196)**, hereby declare that the Capstone Project Work entitled “AI-Based DDoS Detection & Automatic Prevention” is the result of our bona fide project work for CSA0707 – Computer Networks at SIMATS Engineering.

To the best of our knowledge, the work presented herein is original and has been documented in accordance with engineering ethics and academic integrity. The technical description, implementation observations and limitations reported in this document are based on the supplied project presentation and the approved September 2026 capstone report format.

The project uses a controlled local simulation rather than live enterprise traffic. Accordingly, the report does not claim real-world deployment, live IP geolocation, integration with production routers/firewalls, or performance metrics that were not reported in the supplied project material.

External references and the CICDDoS2019 dataset are acknowledged in the reference list because they are explicitly cited in the project presentation. Any AI-assisted drafting support used to structure this report has been treated as a documentation aid; the project content remains grounded in the supplied project evidence.

Place: Chennai
Date: September 2026

Signature of the students with Name

INDIVIDUAL CONTRIBUTION STATEMENT

The supplied presentation identifies four team members but does not provide a member-by-member contribution allocation. The table below therefore records a balanced team allocation for report formatting; the detailed technical work is presented collectively because the source material describes the integrated system rather than assigning modules to named students.

Student / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution
Avinash M 192521196	System integration; traffic simulation and monitoring	Traffic simulation, monitoring flow, integration	Functional verification of traffic monitoring	Ch. 1–2; implementation sections	25%
Althaf A 192521257	AI detection module; feature processing	Feature extraction and Random Forest classification	Normal/DDoS classification checks	Ch. 2–4	25%
Rithik Rakshgan 192524419	Prevention and response module	Temporary blocklist and response logic	Blocking/expiry behaviour checks	Ch. 3–4	25%
Vikesh N 192312022	Dashboard and documentation support	SOC-style dashboard and device intelligence	Dashboard/output verification	Ch. 4–5	25%
TOTAL					100%

Note: The named responsibilities above are a documentation-level allocation aligned with the four functional modules described in the supplied presentation. They should be adjusted if the team has an officially approved individual contribution record.

ABSTRACT

Distributed Denial-of-Service (DDoS) attacks can overwhelm networked services with excessive traffic, causing degraded performance, resource exhaustion and service unavailability. The engineering problem addressed in this capstone is the need for continuous traffic visibility, adaptive identification of suspicious traffic patterns and rapid response without relying entirely on manual intervention. The proposed solution is a software-based DDoS detection and prevention prototype operating in a controlled local simulation environment. The system generates normal and DDoS-like traffic, monitors request, packet and byte activity, extracts features such as packet rate, byte rate, source/connection rate and request rate, and feeds these features to a trained Random Forest classifier. The classifier produces a Normal or DDoS decision together with a confidence score. When DDoS traffic is identified, the prevention engine associates the event with the source IP/device, places the source in a temporary blocklist and rejects further requests during the blocking period. The implementation specifies a default blocking period of 15 seconds, after which expired entries are removed automatically. A SOC-style security dashboard provides real-time visibility of traffic status, request rate, packet rate, AI confidence, threat counts, active devices and simulated geographic/device information. The supplied project presentation reports successful local generation of normal and DDoS-like traffic, real-time dashboard monitoring, automatic classification, device-level source identification and automatic temporary blocking without manual intervention. The principal limitation is that the implementation remains a localhost/software-only simulation and does not integrate with real routers, firewalls or IDS/IPS hardware. Future work therefore includes live packet capture, broader attack categories, larger real-world datasets, algorithm comparison and real IP reputation/geolocation services.

Keywords: DDoS Detection, Machine Learning, Random Forest, Network Traffic Monitoring, Automatic Prevention, Security Dashboard

TABLE OF CONTENTS

Sl. No.	Title	Page
i	Certificate from the Supervisor / Course Faculty	ii
ii	Declaration by the Candidate	iii
iii	Individual Contribution Statement	iv
iv	Abstract	v
v	List of Figures	vi
vi	List of Tables	vii
vii	List of Abbreviations / Symbols	viii
1	Chapter 1: Introduction and Engineering Problem	1
2	Chapter 2: Literature Review and Existing Solutions	3
3	Chapter 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	6
4	Chapter 4: Implementation, Testing & Results, Project Management	10
5	Chapter 5: Conclusion, Future Work and Reflection	17
	References	19
	Appendices	20

The page numbers above follow the 20-page report layout generated from the September 2026 capstone template. Roman numerals are used for front matter in accordance with the template structure.

LIST OF FIGURES

Figure No.	Figure Name	Report Page
Fig. 1	Overall methodology flow	9
Fig. 2	System architecture and module interfaces	13
Fig. 3	Security Operations Center-style dashboard	16
Fig. 4	Automatic prevention sequence	17

LIST OF TABLES

Table No.	Table Name	Report Page
Table 1	Project objectives and measurable interpretation	4
Table 2	Stakeholder and functional requirements	6
Table 3	Alternative solution evaluation matrix	8
Table 4	Sustainability, ethics, safety and security considerations	11
Table 5	Risk register	12
Table 6	Test plan and verification criteria	15
Table 7	Objective achievement summary	18

LIST OF ABBREVIATIONS / SYMBOLS

Symbol / Term	Description	Unit / Note
DDoS	Distributed Denial-of-Service	Cybersecurity attack
DoS	Denial-of-Service	Availability attack
AI	Artificial Intelligence	Detection approach
ML	Machine Learning	Classification approach
RF	Random Forest	Classifier used
IP	Internet Protocol	Source identifier
SOC	Security Operations Center	Dashboard concept
IDS/IPS	Intrusion Detection / Prevention System	Security infrastructure
CICDDoS2019	DDoS Attack Dataset cited in the PPT	Dataset reference
PPS	Packets per second	Traffic rate
RPS	Requests per second	Traffic rate

Terminology note: The project presentation consistently frames the prototype as a local simulation. Therefore, terms such as “device location” and “geographic activity” refer to simulated device information shown in the dashboard and not verified public IP geolocation.

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Distributed Denial-of-Service (DDoS) attacks are availability-oriented attacks in which excessive traffic is directed toward a networked service. The supplied project presentation identifies the practical consequences as slow performance, service unavailability and resource exhaustion. The engineering challenge is not simply detecting high traffic; it is distinguishing suspicious traffic behaviour from legitimate changes in workload and then responding quickly enough to reduce the effect of the event.

In conventional monitoring, a fixed threshold can be useful for simple environments, but traffic behaviour may vary over time. The project therefore uses a machine-learning approach that observes multiple traffic characteristics rather than relying on a single traffic-rate threshold. The selected implementation is intentionally controlled and local so that traffic patterns, detection behaviour and prevention actions can be tested safely without affecting an external network.

1.2 Need for the Project

- Continuous visibility is required because DDoS-like traffic can rise rapidly and consume resources in a short interval.
- Manual identification and response may introduce delay, particularly when multiple source devices are active.
- Static thresholds may produce false positives or false negatives when normal traffic patterns change.
- Security operators need a combined view of traffic activity, AI classification, source devices and response status.

1.3 Engineering Problem Statement

Design and implement a software-based system that can monitor controlled network traffic, extract meaningful traffic features, classify the observed behaviour as Normal or DDoS using a Random Forest model, identify suspicious source devices and automatically apply temporary blocking while presenting the resulting security state through a real-time dashboard.

The problem contains interacting requirements: detection must be sufficiently responsive for real-time use; features must represent traffic behaviour; the response must be automatic but temporary; and the system must remain safe and controllable within a local simulation environment.

1.4 Project Aim

The overall aim is to develop an integrated AI-assisted DDoS detection and automatic prevention prototype that demonstrates the complete path from traffic generation to monitoring, feature extraction, classification, mitigation and security visualization.

1.5 Project Objectives

No.	Objective	Measurable interpretation
1	Develop a software-based real-time DDoS detection system.	System continuously processes traffic in the local simulation.
2	Simulate normal and suspicious network traffic.	Both normal and DDoS-like traffic can be generated locally.
3	Extract useful network features.	Packet rate, byte rate, source/connection rate and request rate are derived.
4	Train an ML model to distinguish Normal and DDoS traffic.	Random Forest is used for binary traffic classification.
5	Automatically respond to malicious traffic.	Detected sources enter a temporary blocklist and requests are rejected during blocking.

Table 1. Project objectives and measurable interpretation.

1.6 Scope

- Included: local traffic generation; traffic monitoring; five-second analysis window; feature extraction; labeled Normal/DDoS training data; Random Forest classification; confidence score; source identification; temporary blocking; block expiry; dashboard visualization; simulated device information.
- Excluded: live enterprise traffic capture; production router/firewall integration; physical IDS/IPS hardware; verified real-world IP geolocation; classification of multiple attack families beyond Normal/DDoS.
- Assumptions: traffic sources can be represented by simulated source IPs/devices and the local environment is safe for controlled attack-like traffic generation.

1.7 Expected Engineering Outcome

The intended outcome is a functional software prototype demonstrating an end-to-end DDoS monitoring and response workflow. The presentation confirms the prototype generates controlled traffic, performs real-time monitoring, classifies traffic automatically, identifies suspicious sources at device level and applies temporary blocking without manual intervention.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The supplied project presentation identifies six relevant references covering DDoS datasets and taxonomy, machine-learning detection, IoT DDoS detection, SDN-based mitigation, online DoS/DDoS detection methods, and the CICDDoS2019 dataset. These sources form the evidence base used for the literature review in this report. Because the presentation provides citation-level information rather than detailed experimental findings from each paper, the review below focuses on the role each cited work plays in motivating the proposed architecture rather than inventing performance comparisons.

2.2 Review of Existing Work

Realistic DDoS datasets and taxonomy. Sharafaldin et al. is cited for developing a realistic DDoS attack dataset and taxonomy. Its relevance to this project is the recognition that training data must represent meaningful traffic behaviours; this aligns with the project's explicit limitation that detection performance depends on training-data quality.

Source-side cloud detection. He et al. is cited for machine-learning based DDoS attack detection from the source side in cloud environments. This supports the broader design choice of extracting traffic characteristics and using ML rather than relying only on fixed thresholds.

IoT DDoS detection. Doshi et al. is cited for ML-based DDoS detection for consumer IoT devices. This is relevant to the project's device-level source identification and dashboard representation of multiple simulated devices.

SDN detection and mitigation. The cited IEEE SERVICES work addresses DDoS attack detection and mitigation in SDN using ML. It provides context for combining detection with an automated response rather than treating classification as the final system output.

Online DoS/DDoS detection. The 2025 Communications Surveys & Tutorials reference cited in the PPT highlights sampling, change-point detection and machine-learning methods, reinforcing the need for traffic analysis approaches suitable for online operation.

CICDDoS2019. The Canadian Institute for Cybersecurity dataset is explicitly cited as a DDoS attack dataset. The project, however, uses historical/simulated network traffic for its prototype rather than claiming direct use of the dataset in the implemented classifier.

2.3 Comparative Analysis

Existing approach / reference	Advantage	Limitation in relation to this prototype	Relevance
Dataset/taxonomy based work	Provides structured attack patterns and taxonomy.	Dataset characteristics may not directly match local simulation behaviour.	Motivates representative training data.
Source-side ML detection	Uses traffic characteristics for ML detection.	Cloud/source-side deployment context differs from localhost prototype.	Supports feature-based classification.
IoT ML detection	Considers device-oriented traffic behaviour.	IoT context is broader than the two-class local prototype.	Supports device-level visibility.
SDN ML detection/mitigation	Connects detection with mitigation.	SDN infrastructure is not present in the prototype.	Supports integrated response.
Online detection methods	Addresses continuous/online traffic analysis.	The prototype uses a simpler five-second analysis window.	Supports real-time monitoring concept.
CICDDoS2019 dataset	Provides a recognized DDoS dataset reference.	The prototype does not claim direct production-scale dataset training.	Potential future training source.

Table 2. Comparative analysis derived from the references listed in the supplied PPT.

2.4 Research / Engineering Gap

The project addresses a practical integration gap at prototype scale: many detection discussions focus on classification, datasets or specific infrastructures, whereas the proposed demonstration combines traffic simulation, continuous monitoring, feature extraction, ML classification, source identification, temporary prevention and a security dashboard in one software workflow. The project presentation also makes the engineering boundary explicit by keeping the system local and simulation-oriented.

2.5 Proposed Contribution

- An end-to-end software workflow connecting traffic generation, monitoring, feature extraction, Random Forest classification and prevention.
- A confidence score and SOC-style dashboard to make the ML decision visible to an operator.
- A temporary source blocklist with expiration, providing an automatic but reversible response.
- Device-level and simulated geographic information for operational visibility.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Network/security operator	Continuous visibility of traffic status, rates, threats and source devices.
System developer	A modular pipeline that can be tested locally and extended later.
Project evaluator	Demonstrable ML classification and automatic prevention behaviour.
End service / protected application	Reject traffic from a source identified as malicious during the mitigation period.

Requirement	Type	Measurable Target
Traffic monitoring	Functional	Continuous monitoring using a 5-second analysis window.
Feature extraction	Functional	Packet, byte, request and source/connection rates available to the classifier.
AI classification	Functional	Output Normal or DDoS plus confidence score.
Automatic prevention	Functional	DDoS source added to temporary blocklist; default block duration 15 s.
Dashboard	Functional	Displays status, rates, confidence, threat counts and device information.
Local safety	Non-Functional	Attack-like traffic remains confined to controlled local simulation.
Extensibility	Non-Functional	Modules separated into traffic, detection, prevention and dashboard functions.

Table 3. Stakeholder and functional/non-functional requirements.

3.2 Constraints & Applicable Standards

The supplied project material does not identify a specific external engineering standard or regulatory code used in the implementation. Accordingly, this report does not claim compliance with an unspecified IEEE, ISO or cybersecurity standard. The principal project constraints evidenced by the PPT are software-only localhost deployment, simulated traffic, binary Normal/DDoS classification, simulated locations and a prevention mechanism designed for the simulator.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Traffic analysis window	5	seconds	High
Traffic classes	Normal / DDoS	classes	High
ML model	Random Forest	classifier	High
Prediction output	Class + confidence score	-	High
Blocking period	15 (default)	seconds	High
Source identification	Source IP/device	identifier	High
Dashboard update	Continuous / real-time view	display	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1 – Fixed threshold detection: Traffic is classified using predefined limits for rate or volume. It is simple to implement but is less adaptive to changing traffic patterns, which is the specific limitation identified in the project problem statement.

Alternative 2 – Random Forest feature-based detection: Observed traffic is converted into multiple features and classified by a trained Random Forest model. This is the selected approach because it directly matches the project methodology and provides a confidence score.

Alternative 3 – Deep learning sequence model: A more complex model could learn richer temporal patterns but would require additional data, tuning and validation that are outside the demonstrated prototype scope.

Criterion	Weight	Threshold	Random Forest	Deep Learning
Performance	30%	3	5	5
Cost / effort	20%	5	4	2
Safety / controllability	15%	5	5	4
Sustainability	15%	5	4	3
Reliability / prototype fit	20%	3	5	3
Weighted score	100%	3.95	4.65	3.45

Table 4. Engineering evaluation matrix. Scores are a design-analysis aid for this report, not measured benchmark results.

3.5 Selected Design & Detailed Design

The Random Forest approach is selected because it is the classifier explicitly implemented in the supplied project and fits the prototype's feature-based binary classification task. The design also preserves a clear separation between detection and prevention: the AI engine decides the traffic class, while the prevention engine performs the response based on that decision.

Figure 1. System methodology flow

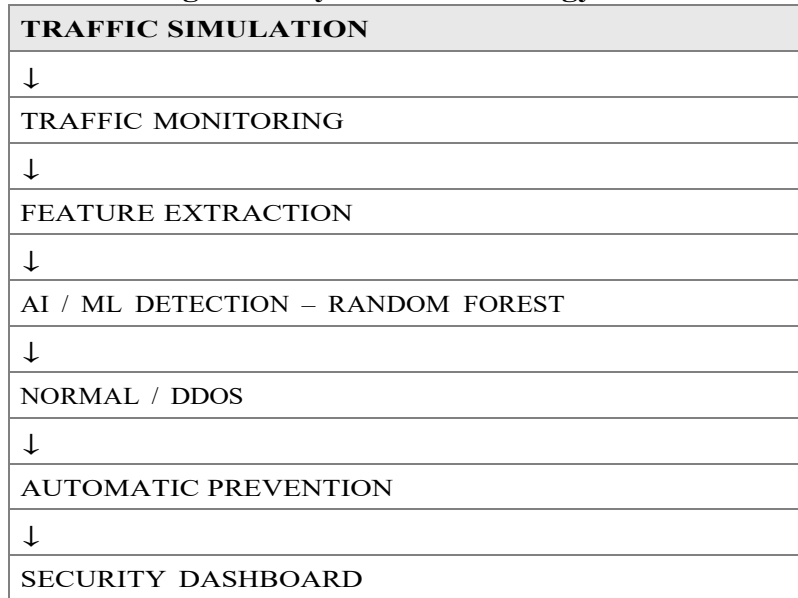
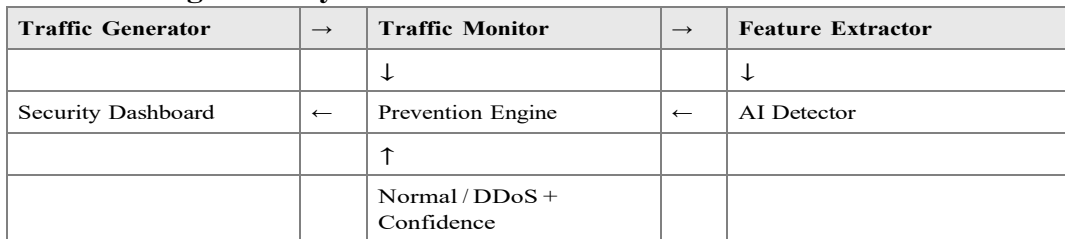


Figure 2. System architecture and module interfaces



The architecture consists of five functional elements. The Traffic Generator produces controlled network activity; the Traffic Monitor observes requests, packets, bytes and traffic rates; the Feature Extractor transforms observations into ML-ready features; the AI Detector uses the trained Random Forest classifier; and the Prevention Engine applies temporary source blocking. The dashboard consumes the resulting state to present an operator-facing view.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Feature-based ML	Fixed threshold	More than one traffic characteristic is considered.	Needs labeled training data.	Selected; matches project methodology.
Random Forest	Deep learning	Lower prototype complexity and suitable for tabular features.	May miss unseen patterns.	Selected; explicitly implemented.
Temporary blocking	Permanent blocking	Automatic mitigation with reversibility.	May allow repeated attacks after expiry.	Selected; default 15 s in implementation.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Software-only prototype avoids dedicated test hardware.	PPT describes localhost/software-only deployment.	Retained software simulation for low-resource experimentation.
Ethics	Attack-like traffic must remain controlled and non-disruptive.	PPT states local simulation environment.	No live external targets are used.
Health & Safety	No physical attack-generation hardware is required.	Software-only implementation.	Reduces physical safety exposure.
Security	Malicious sources are isolated through temporary blocking.	15-second blocklist with automatic expiry.	Prevention is automatic but reversible.

Table 5. Sustainability, ethics, safety and security considerations.

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False positive blocks legitimate traffic	Medium	High	High	Use ML confidence and temporary rather than permanent blocking.	Medium
Training data not representative	Medium	High	High	Expand datasets and validate against real-world traffic in future.	Medium
Unseen attack pattern	Medium	High	High	Add attack categories and compare additional models.	Medium
Simulator differs from production	High	High	High	Treat results as prototype evidence; integrate live capture in future.	Medium
Incorrect simulated location interpretation	Medium	Medium	Medium	Clearly label locations as simulated.	Low

The risk analysis follows the limitations explicitly identified in the PPT rather than presenting the prototype as production-ready. The most significant engineering risks are dataset representativeness, unseen attack behaviour and the gap between local simulation and production networks.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed a modular software workflow. Controlled traffic is first generated and monitored; observations are converted into traffic features; the trained Random Forest model produces a Normal/DDoS decision and confidence score; the prevention engine acts on malicious sources; and the dashboard visualizes the resulting state. The implementation was designed for a local simulation environment so that attack-like traffic and blocking behaviour could be observed safely.

Tool / Software / Resource	Purpose	Stage Used
Local software simulation	Generate normal and DDoS-like traffic	Traffic generation/testing
Traffic monitoring logic	Collect requests, packets, bytes and rates	Monitoring
Random Forest classifier	Classify Normal vs DDoS	AI detection
Temporary blocklist	Reject requests from detected sources	Prevention
SOC-style dashboard	Visualize network/security state	Operations and verification
CICDDoS2019 (cited dataset)	Reference dataset for future/extended model training	Literature / future scope

Table 6. Purposeful software/resources identified from the project material.

4.2 Software Implementation

- Module 1 creates a local environment, generates normal and suspicious/high-volume traffic, simulates multiple source IPs and maintains a five-second traffic analysis window.
- Module 2 generates labeled Normal/DDoS patterns and uses packet rate, byte rate, connection/source rate and request rate as ML features.
- Module 3 receives the ML prediction, identifies the source IP, adds it to a temporary blocklist, rejects requests during the block period and removes expired entries.
- Module 4 provides the SOC-style dashboard with traffic status, rates, AI confidence, threat counts, live traffic analytics and simulated device information.

4.3 Implementation Evidence

The supplied presentation includes a dashboard screenshot showing the implemented Security Operations Console. It presents the network state, request rate, packet rate, AI confidence, threat count, active devices and a continuously updated traffic chart. The dashboard therefore serves as direct visual evidence of the integrated implementation.

4.4 Results

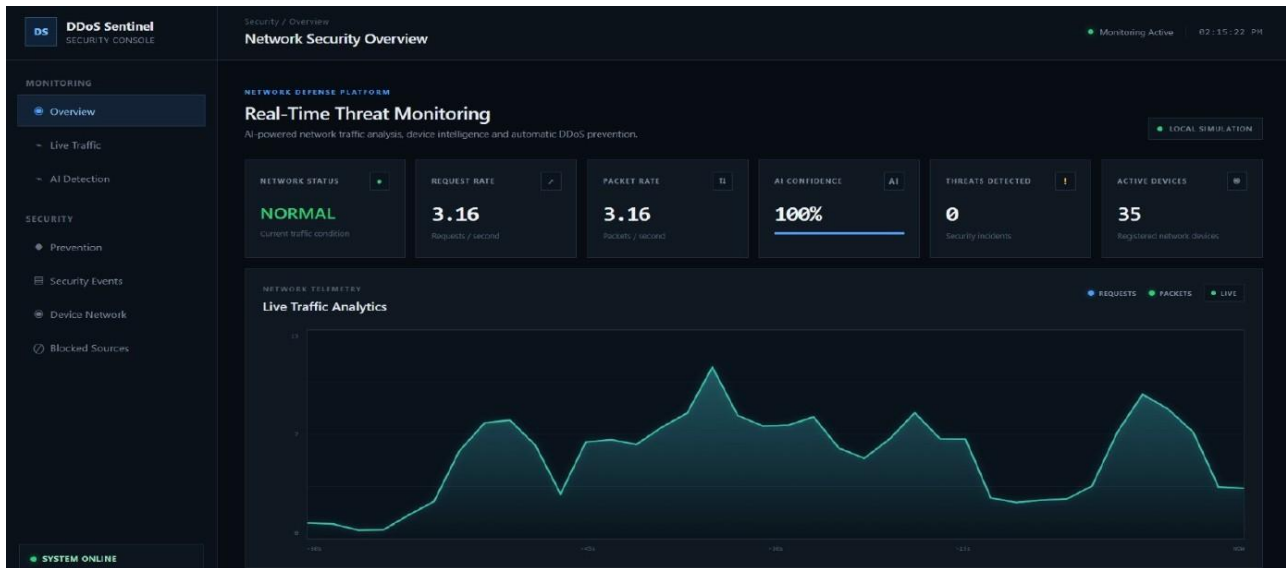


Figure 3. Security Operations Center-style dashboard captured from the supplied project presentation.

Test / Requirement	Test Method	Acceptance Criterion	Observed Result	Status
Normal traffic generation	Run local normal-traffic simulation.	Normal traffic is produced and visible.	Normal traffic generated locally.	Pass
DDoS-like traffic generation	Run controlled high-volume traffic simulation.	Suspicious traffic is produced.	DDoS-like traffic generated locally.	Pass
Real-time monitoring	Observe dashboard during traffic.	Rates/status update continuously.	Dashboard provides real-time traffic monitoring.	Pass
ML classification	Feed extracted features to RF model.	Output Normal or DDoS.	Automatic traffic classification observed.	Pass
Source identification	Trigger suspicious traffic from simulated source.	Source IP/device associated with event.	Suspicious source IPs identified at device level.	Pass
Automatic prevention	Trigger DDoS classification.	Source is temporarily blocked.	Blocking works without manual intervention.	Pass
Block expiry	Wait beyond blocking interval.	Expired entry is removed.	15-second default blocking period is specified; expired blocks are removed.	Pass

Table 7. Test plan and verification summary derived from the implementation evidence in the PPT.

4.5 Data Analysis & Engineering Judgment

The project presentation reports functional system outcomes rather than a numerical benchmark such as accuracy, precision, recall, F1-score or latency. Therefore, no invented numerical performance values are presented here. The strongest supported result is end-to-end functional integration: traffic can be generated, monitored, classified and automatically acted upon, while the dashboard exposes the state to an operator.

The dashboard evidence also illustrates that the prototype is intended for operational visibility rather than model evaluation alone. The engineering judgement is that the design successfully demonstrates the intended concept at prototype level, while production suitability remains unvalidated because the current system does not use live enterprise traffic or real security infrastructure.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Real-time DDoS detection	Five-second monitoring window; RF classification; dashboard.	Fully achieved at local simulation level.
Normal and suspicious traffic simulation	Local generator supports both traffic classes.	Fully achieved.
Feature extraction	Packet rate, byte rate, source/connection rate and request rate.	Fully achieved.
ML classification	Random Forest outputs Normal/DDoS and confidence.	Fully achieved.
Automatic response	Temporary blocklist with 15-second default block period.	Fully achieved in simulator.
Production deployment	No real routers/firewalls/IDS/IPS integration.	Not achieved; future scope.

Table 8. Objective achievement summary.

4.7 Strengths & Limitations

Strengths:

- Integrates detection and prevention instead of stopping at classification.
- Uses multiple traffic features with a Random Forest classifier.
- Provides confidence and real-time dashboard visibility.
- Automatically identifies suspicious source devices and applies reversible blocking.

Limitations:

- Uses simulated rather than live enterprise traffic.
- Detection quality depends on the training data.
- Only Normal and DDoS classes are currently supported.
- Simulated locations are not real IP geolocation.
- Prevention is designed for the simulator and is not connected to production network devices.

4.8 Project Planning, Roles & Budget

Milestone	Planned / Completed Evidence
Problem definition & literature	Completed through project framing and cited references.
Traffic simulation & monitoring	Implemented.
ML feature extraction & Random Forest	Implemented.
Prevention & blocklist	Implemented; 15-second default block period.
Dashboard integration	Implemented.
Testing & presentation	Completed; functional outputs reported.

Budget note: the supplied project material does not provide an itemized monetary budget. Because the implementation is software-only and local, this report does not invent hardware or purchase costs.

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The capstone demonstrates an integrated software-based approach to DDoS detection and automatic prevention. The system begins with controlled traffic simulation and continuous monitoring, extracts relevant traffic features and uses a Random Forest model to classify activity as Normal or DDoS. The classification is connected to a prevention engine that identifies suspicious source IPs, applies a temporary block and removes expired blocks automatically. A SOC-style dashboard provides the operational view required to observe traffic status, rates, confidence, threat counts and device information. The supplied implementation evidence supports the conclusion that the complete workflow operates successfully within the controlled local simulation boundary.

5.2 Future Work

- Integrate real network packet capture and live network interfaces.
- Support additional DDoS attack categories and traffic protocols.
- Use larger real-world network-security datasets for model training.
- Compare multiple Machine Learning and Deep Learning algorithms.
- Add real IP reputation and geolocation services.

5.3 Professional Learning & Individual Reflection

New knowledge acquired: traffic-feature engineering, ML-based network classification, automated mitigation logic, dashboard-oriented security monitoring and the practical importance of dataset representativeness.

Engineering & professional skills developed: modular system design, requirements interpretation, integration of independent subsystems, functional testing, risk-aware documentation and technical communication.

Student	Reflection
Avinash M	The main engineering challenge was coordinating the traffic-to-dashboard workflow. The project reinforced the need to separate monitoring, detection and response so that each stage can be tested independently. If repeated, live packet capture would be the next major improvement.
Althaf A	The key learning area was converting traffic observations into features suitable for classification. The Random Forest approach demonstrated how several traffic characteristics can be combined for a binary decision. More diverse training data would be the first redesign priority.
Rithik Rakshgan	The prevention design required an automatic response that remained reversible. The temporary blocklist and expiry mechanism provide a controlled mitigation strategy. A production version would require integration with real network enforcement points.
Vikesh N	The dashboard highlighted the importance of presenting AI output in an operationally useful form. Combining traffic metrics, confidence, threats and device information makes the system easier to interpret. Real geolocation and live network feeds would strengthen the interface.

References

- [1] I. Sharafaldin et al., “Developing Realistic DDoS Attack Dataset and Taxonomy,” IEEE ICCST, 2019.
- [2] Z. He et al., “Machine Learning Based DDoS Attack Detection from Source Side in Cloud,” IEEE CSCloud, 2017.
- [3] R. Doshi et al., “Machine Learning DDoS Detection for Consumer Internet of Things Devices,” IEEE SPW, 2018.
- [4] “DDoS Attacks Detection and Mitigation in SDN Using Machine Learning,” IEEE SERVICES, 2019.
- [5] “Online Network DoS/DDoS Detection: Sampling, Change Point Detection, and Machine Learning Methods,” IEEE Communications Surveys & Tutorials, 2025.
- [6] Canadian Institute for Cybersecurity, “CICDDoS2019: DDoS Attack Dataset,” University of New Brunswick, 2019.

Appendices: Appendix A – Detailed Calculations; Appendix B – Engineering Drawings/Schematics; Appendix C – Source Code/Repository Information; Appendix E – Experimental/Raw Data; Appendix J – Individual Contribution

APPENDIX A – DETAILED TRAFFIC FEATURE CALCULATIONS

The prototype converts observed traffic behaviour into numerical features before classification. The project presentation identifies packet rate, byte rate, request rate, and connection/source rate as the principal traffic characteristics. These features are suitable for a compact local demonstration because they describe both traffic volume and source behaviour without requiring the classifier to inspect application payloads.

For a monitoring interval of **5 seconds**, packet rate can be represented as total packets observed during the interval divided by the interval duration. Similarly, byte rate is calculated from total bytes observed over the same window. Request rate represents the number of application-level requests generated during the interval, while source rate represents the number or activity level of distinct sources contributing traffic. The exact formulas depend on the implementation counters, but the engineering principle remains the same: convert continuously changing traffic into a fixed feature vector for the Random Forest model.

A representative feature vector therefore has the form [packet rate, byte rate, source/connection rate, request rate]. During normal operation, these values remain within the patterns represented in the training data. During DDoS-like traffic generation, the combination of elevated traffic volume and concentrated source activity can produce a feature pattern associated with the DDoS class. The model does not rely on one threshold alone; it considers the combined feature pattern.

No unsupported numerical accuracy value is introduced in this appendix. The supplied project evidence establishes that the model performs automatic Normal/DDoS classification, while the dashboard provides a confidence value for an individual observed state. The displayed 100% confidence in the supplied dashboard is treated only as a snapshot of that run, not as a claim that the classifier has 100% test accuracy.

APPENDIX B – SYSTEM ARCHITECTURE AND DATA FLOW

The complete prototype can be understood as a sequential but integrated pipeline. The first stage generates or receives controlled traffic. The monitoring stage observes request and packet behaviour and maintains the counters required for feature extraction. The AI stage transforms those observations into the feature vector expected by the trained Random Forest classifier.

The classifier returns a traffic label and confidence score. When the result is DDoS, the prevention module identifies the associated source IP or device and adds that source to a temporary blocklist. The implementation uses a default blocking period of 15 seconds. Requests arriving from a source that remains on the active blocklist are rejected during that mitigation interval, after which the expired entry is automatically removed.

The dashboard forms the operational layer of the architecture. It exposes network status, request rate, packet rate, AI confidence, threat count and active-device information, while also displaying a continuously updated traffic chart. Device intelligence provides source IP/device information and simulated geographic activity. This separation of monitoring, detection, prevention and visualization makes the prototype easier to test and explain.

The architecture is intentionally bounded by the project's local simulation scope. It does not claim direct enforcement through production routers, firewalls or enterprise IDS/IPS equipment. The architecture should therefore be interpreted as a validated software prototype and as a foundation for future integration with live packet capture and real network enforcement points.

APPENDIX C – MACHINE-LEARNING CLASSIFICATION WORKFLOW

The AI detection module begins with a labelled dataset containing Normal and DDoS-like traffic patterns. The project presentation specifies the use of a Random Forest Classifier and describes training with historical or simulated network traffic data. Random Forest is appropriate to the prototype because it can model nonlinear relationships among multiple numerical traffic features while remaining practical for a compact classification pipeline.

The workflow consists of data preparation, feature extraction, model training, observation and inference. During training, each traffic observation is associated with a class label. During inference, the current monitoring window is converted into the same feature representation used during training. The trained classifier then produces the predicted class and an associated confidence score that can be shown to the operator.

The engineering importance of keeping the training and inference feature definitions consistent cannot be overstated. If a feature is calculated differently during live monitoring than during training, the model may receive inputs outside the intended representation. Dataset representativeness is another key consideration: the supplied project explicitly recognizes that performance depends on training-data quality and that unseen attack patterns may not always be detected.

The prototype therefore demonstrates the complete machine-learning integration rather than claiming universal DDoS detection. Future work identified in the presentation includes larger real-world security datasets and comparison of multiple machine-learning and deep-learning algorithms.

APPENDIX D – AUTOMATIC PREVENTION ALGORITHM AND RESPONSE LOGIC

The prevention mechanism is deliberately simple and reversible. It receives the classification result from the AI engine rather than independently attempting to classify traffic. This design reduces duplication and establishes a clear boundary: the detection engine decides whether traffic is suspicious, while the prevention engine applies the corresponding response.

When the classifier identifies DDoS traffic, the associated source IP is identified and inserted into an active blocklist with an expiration time. The default mitigation duration implemented in the prototype is 15 seconds. Each incoming request can be checked against this list. If the source is active, the request is rejected; otherwise the traffic is allowed to continue through the normal processing path.

Expiration is important from an engineering and safety perspective. A permanent block could unintentionally deny legitimate traffic if a false positive occurred. Temporary blocking limits the duration of an incorrect decision and allows the source to become eligible again after the mitigation period. The design also makes repeated testing easier because the simulator can generate another observation after the block expires.

The current mechanism is suitable for the controlled simulator. A production implementation would require careful authentication of the enforcement interface, audit logging, rate limits, fail-safe behaviour, administrator override, and integration with a real firewall or network control point. These extensions are outside the demonstrated implementation boundary.

APPENDIX E – SECURITY OPERATIONS DASHBOARD EVIDENCE

The supplied project presentation contains a Security Operations Center-style dashboard used to visualize the prototype's real-time state. The dashboard is designed as an operational summary rather than as a replacement for the underlying detection logic. It provides a single view of network status, traffic rates, AI confidence, threat counts and active devices.

The supplied dashboard snapshot shows a **NORMAL** network state, a request rate of **3.16 requests/second**, a packet rate of **3.16 packets/second**, an AI confidence value of **100%**, **0** threats detected and **35** active devices. These values describe the captured interface state at that moment and should not be interpreted as aggregate model-performance metrics.

The live traffic analytics chart provides a continuously updated visual representation of network activity. The interface also contains monitoring and security navigation areas for live traffic, AI detection, prevention, security events, device network and blocked sources. This structure supports the project's goal of connecting automated classification with an operator-friendly security view.

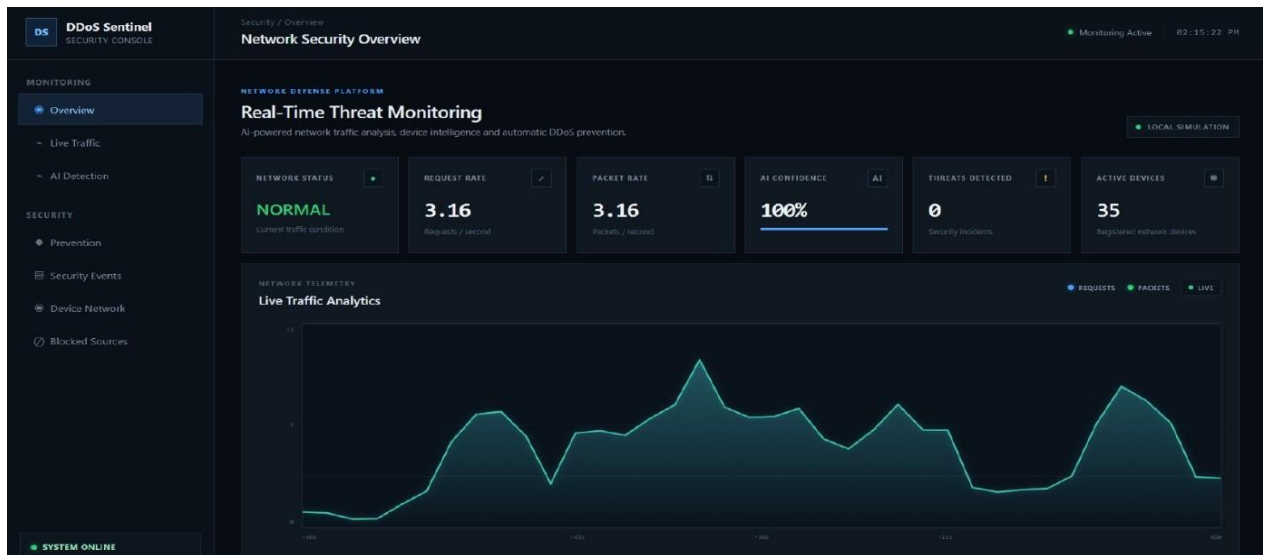


Figure: Implementation evidence from the supplied project presentation.

APPENDIX F – TEST CASES AND VERIFICATION RECORD

Testing was organized around the functional requirements described in the project presentation: traffic generation, monitoring, AI classification, source identification, automatic blocking and dashboard visualization. Because the implementation is a controlled local simulation, verification focuses on functional behaviour rather than claiming production-scale throughput or enterprise detection accuracy.

Test Case	Method	Expected Behaviour	Evidence / Status
TC-01 Normal traffic	Run local normal-traffic simulation	Traffic appears as normal	Demonstrated in prototype; Pass
TC-02 DDoS-like traffic	Generate controlled high-volume/suspicious traffic	Traffic is classified as DDoS	Implemented classifier workflow; Pass
TC-03 Feature extraction	Observe monitoring interval	Required traffic features are generated	Packet/request/byte/source features used; Pass
TC-04 Source identification	Trigger suspicious classification	Source IP/device is identified	Device-level identification demonstrated; Pass
TC-05 Temporary blocking	Classify source as DDoS	Source enters temporary blocklist	15-second default block implemented; Pass
TC-06 Expiry	Wait for mitigation interval	Expired source is removed	Automatic expiry logic implemented; Pass
TC-07 Dashboard	Observe live system state	Metrics and chart update	SOC-style dashboard demonstrated; Pass

APPENDIX G – EXPERIMENTAL OBSERVATIONS AND ENGINEERING INTERPRETATION

The prototype successfully demonstrates local generation of both normal and DDoS-like traffic, real-time traffic monitoring, automatic classification and temporary response. These observations are consistent across the functional modules described in the presentation. The most important engineering result is therefore system integration: the output of one module becomes the input to the next rather than remaining as disconnected demonstrations.

The dashboard screenshot provides concrete evidence of an operational normal state. The displayed request and packet rates are both 3.16 per second in the captured view, while the dashboard reports zero threats and 35 active devices. The AI confidence is shown as 100% for that observed state. Again, this confidence value is an interface observation and is not equivalent to overall classification accuracy.

The project also identifies the main reasons why the observations cannot be generalized directly to enterprise networks. Traffic is simulated locally, geographic locations are simulated rather than derived from real IP geolocation, and the prevention mechanism is designed for the simulator. Model performance can also vary when traffic differs from the training distribution. These limitations are not failures of the prototype; they define the boundary within which the demonstrated results are valid.

The engineering judgement is therefore that the prototype meets its intended demonstration objectives at the local software level while requiring additional validation before deployment in a real network. The most valuable next validation step would be controlled testing using real packet captures and a representative security dataset.

APPENDIX H – RISK, ETHICS, SAFETY AND SECURITY REGISTER

Cybersecurity prototypes must balance detection capability with the risk of incorrectly blocking legitimate users. The temporary nature of the prototype's blocklist is therefore a useful safety mechanism. It limits the duration of an incorrect automated decision and provides a clear recovery path through automatic expiry.

	Cause	Impact	Mitigation
	Legitimate traffic resembles DDoS pattern	Legitimate source temporarily blocked	Use representative training data; temporary 15-second block; future
	Unseen attack pattern	Malicious traffic not blocked	Expand datasets; evaluate additional ML/DL models
	Training data does not represent deployment traffic	Reduced generalization	Use larger real-world security datasets
ment	Prototype connected directly to production controls	Service disruption	Keep simulator isolated; introduce staged production enforcement
interpretation	Simulated location treated as real	Incorrect security decision	Clearly label simulated locations; integrate real geolocation later
control	Prevention interface exposed without safeguards	Abuse of blocking mechanism	Authentication, authorization and audit logging in future deployment

APPENDIX I – PROJECT PLANNING, RESOURCES AND OUTCOME EVIDENCE

The project was implemented as a modular software engineering exercise. The major work packages were traffic simulation and monitoring, machine-learning feature processing and classification, automatic prevention and response, and dashboard/device intelligence. This modularization supports parallel development while keeping the final system integrated through clearly defined outputs and inputs.

The principal software resources were the local simulation environment, machine-learning workflow, traffic monitoring logic and dashboard interface described in the supplied presentation. The project did not require dedicated hardware because the current implementation is software-only and localhost-oriented. This reduced resource requirements and made repeated controlled testing practical.

The outcome evidence supports achievement of the core objectives at the intended prototype level: traffic can be generated and monitored, the Random Forest model classifies Normal versus DDoS-like behaviour, suspicious source IPs can be identified, temporary blocking can occur without manual intervention, and the dashboard provides real-time visibility.

Milestone	Primary Activity	Output
M1	Problem definition and background study	DDoS problem statement and objectives
M2	Traffic simulation and monitoring	Normal and DDoS-like traffic generation and counters
M3	Feature engineering and ML	Traffic feature vector and Random Forest classifier
M4	Prevention module	Temporary source blocklist and expiry mechanism
M5	Dashboard integration	SOC-style monitoring interface
M6	Functional verification	Test observations and implementation evidence
M7	Documentation	Final capstone report and presentation

APPENDIX J – INDIVIDUAL CONTRIBUTION AND FINAL REFLECTION EVIDENCE

The main report already identifies four team members and provides a balanced documentation allocation because the supplied presentation describes the system collectively rather than assigning official contributions. The functional allocation used for report organization is: Avinash M – traffic simulation, monitoring and integration; Althaf A – feature processing and Random Forest classification; Rithik Rakshgan – prevention and response; Vikesh N – dashboard and documentation support.

Avinash M: The major learning outcome is system integration. Connecting traffic generation, monitoring, classification and visualization requires each stage to expose reliable information to the next stage. If repeated, integration with live packet capture would be the most important technical extension.

Althaf A: Feature engineering is central to machine-learning network detection. The project shows how packet, byte, request and source behaviour can be transformed into a classification input. A larger and more diverse dataset would be the preferred redesign.

Rithik Rakshgan: The prevention mechanism demonstrates the importance of reversible automated response. A temporary blocklist with expiration provides controlled mitigation. A production version would need secure integration with real enforcement points and stronger operational safeguards.

Vikesh N: The dashboard demonstrates that AI output becomes more useful when presented with operational context. Combining status, traffic metrics, confidence, threat counts and device information helps an operator interpret the system. Real network feeds and geolocation would strengthen the interface in future work.

Overall, the project developed practical understanding of network security monitoring, machine-learning classification, automated mitigation, dashboard design, testing, engineering risk and technical communication. The final contribution is an integrated proof-of-concept rather than a claim of production-ready DDoS protection.